DAMBI DOLLO UNIVERSITY

Research, Publication, and Ethics Directorate Office

Concept Note / Synopsis

THEME 2: Science, Technology, and Innovation

Sub-Theme: AI and Machine Learning

TITLE:

Development and Evaluation of a Deep Learning-Based Mobile Decision Support System for Early Detection and Management of Coffee Diseases among Smallholder Farmers in Western Oromia, Ethiopia.

Authors:

No	Name	College	Department	Specialization	Affiliation
1	Geleta Tilahun	CoET	Information Technology	Computer Science and Engineering	DaDU
2	Abdi Kenesa	CoNS	Statistics	Biostatistics	DaDU
3	Zelalem Teso	CoANR	Horticulture	Horticulture	DaDU

1. Background

Coffee is one of Ethiopia's important agricultural products and provides income and livelihoods for many smallholder farmers, particularly in Western Oromia. However, coffee production is affected by diseases that can reduce plant health, productivity, and coffee quality. Early and accurate disease identification is essential for effective management, but farmers often depend on visual observation and limited access to agricultural extension workers or plant health experts. Recent advances in artificial intelligence, particularly deep learning and computer vision, provide an opportunity to identify coffee diseases from plant images captured using widely available smartphones.

However, AI models developed using controlled or geographically different datasets may not perform reliably under the diverse conditions found in farmers' fields, where lighting, background, camera quality, leaf orientation, and disease severity vary. Therefore, this research proposes to develop and evaluate a deep-learning-based mobile system using coffee disease images collected under real field conditions in Western Oromia. The system will support farmers and agricultural extension workers in identifying major coffee diseases and accessing appropriate management information, with Afaan Oromoo included where appropriate to improve accessibility, understanding, and practical use.

2. Problem Statement

Coffee diseases can cause serious losses when they are not identified and managed early. Farmers may delay appropriate action or apply unsuitable management practices when they cannot correctly recognize disease symptoms, allowing diseases to spread and reducing productivity. Although previous studies in Ethiopia and other coffee-producing countries have demonstrated the potential of machine learning and deep learning for identifying coffee diseases from images, many have focused mainly on model development and classification using specific datasets. Their practical application under real farming conditions, particularly using locally collected images, remains limited.

Therefore, there is a need to develop and evaluate a deep-learning model using coffee images collected under realistic field conditions in Western Oromia and determine its reliability for disease identification. The study will also consider developing a lightweight

model suitable for smartphones and presenting disease information in a clear form for farmers. By integrating agriculture, computer science, deep learning, data analysis, mobile technology, and agricultural extension, the proposed study aims to bridge the gap between AI-based disease classification and practical coffee disease management.

3. General Objective

To develop and evaluate a deep learning based mobile decision support system for early detection and management of major coffee diseases among smallholder farmers in Western Oromia, Ethiopia.

4. Specific Objectives

- To identify major coffee diseases relevant to the selected study areas.
- To collect and develop a locally representative image dataset of healthy and diseased coffee plants under field conditions.
- To develop and compare deep-learning models for classification of major coffee diseases.
- To identify a suitable lightweight model for smartphone-based disease detection.
- To develop a prototype mobile application that provides coffee disease identification and related management information.
- To provide clear disease information and management recommendations in Afaan Oromoo.
- To evaluate the accuracy, usability, and practical usefulness of the developed system with farmers and agricultural extension workers.

5. Research Questions

1. How accurately can deep-learning models identify major coffee diseases using smartphone images collected under field conditions in Western Oromia?
2. Which deep-learning model provides an appropriate balance between classification accuracy and computational requirements for smartphone deployment?
3. How does a model trained or fine-tuned using locally collected coffee images perform under actual field conditions?
4. How useful is the proposed system for supporting preliminary coffee disease identification and management decisions?
5. How do farmers and agricultural extension workers perceive the usability and usefulness of the proposed mobile system?

6. Methodology

6.1. Study Area and Research Design

The study will be conducted in selected coffee producing areas of Western Oromia. The final study sites will be selected in consultation with relevant agricultural offices based on coffee production, disease occurrence, accessibility, and the feasibility of working with farmers and extension personnel. An applied and experimental research design will be used. The study will combine field data collection, agricultural disease identification, image processing, deep-learning model development, mobile application development, and user evaluation.

6.2. Identification of Coffee Diseases

The diseases included in the study will be determined after consultation with plant pathologists, agricultural experts, extension workers, and field assessment. This approach will ensure that the selected disease classes reflect diseases that are relevant to the study area rather than simply adopting disease categories from an external dataset. Coffee disease images will be reviewed and labeled with support from qualified agricultural or plant health experts.

6.3. Image Data Collection

Images of healthy and diseased coffee plants will be collected using smartphones from selected farms. The researchers will intentionally include photographs taken under normal field conditions rather than relying only on carefully controlled images.

The dataset will therefore attempt to capture variation in:

- lighting and shadows;
- background conditions;
- camera quality;
- leaf orientation;
- disease stage;
- disease severity; and
- partial or unclear symptoms.

The images will be checked and labeled before being used for model training and evaluation. Appropriate training, validation, and testing procedures will be followed to reduce the risk of overestimating model performance.

6.4. Deep Learning Model Development

Several suitable deep-learning architectures will be investigated. Depending on the characteristics of the dataset, candidate models may include CNN-based architectures such as ResNet, DenseNet, EfficientNet, and MobileNet. Transfer learning will also be considered because it can be useful when the locally collected dataset is relatively limited. The models will be compared using:

- accuracy;
- precision;
- recall;
- F1-score;
- confusion matrix;
- model size; and
- inference time.

The final model will be selected not only according to classification performance but also according to its suitability for practical smartphone deployment.

6.5. Mobile Application Development

A prototype mobile application will be developed to allow a user to take or upload a photograph of a coffee plant. The application will process the image and provide the predicted disease class together with relevant information about the disease and appropriate management recommendations. The research will investigate the possibility of running the trained model directly on a smartphone so that basic disease detection can be available even where Internet connectivity is limited.

6.6. Afaan Oromoo Information

The system will include Afaan Oromoo information where appropriate. Agricultural and plant-health experts will help ensure that disease descriptions and management information are technically appropriate, while Afaan Oromoo expertise will help make the information understandable to farmers.

6.7. User Evaluation

The prototype will be evaluated with selected farmers and agricultural extension workers.

The evaluation will consider:

- ease of use;
- clarity of the information provided;
- perceived usefulness;
- response time;
- confidence in using the system; and
- usefulness for preliminary disease identification.

The evaluation will complement the technical assessment of the deep-learning models and provide evidence about the practicality of the proposed technology.

7. Expected Outputs

The study is expected to produce:

1. A locally collected and expert-validated coffee disease image dataset.
2. One or more evaluated deep-learning models for coffee disease classification.

3. A lightweight model suitable for investigation of smartphone deployment.
4. A prototype mobile coffee disease detection and decision-support application.
5. Coffee disease information and management content in Afaan Oromoo.
6. Technical and user-evaluation results from the study area.
7. A foundation for future research on AI-supported coffee disease surveillance.

8. Expected Community Benefit

The main intended benefit is to improve access to preliminary coffee disease information for smallholder farmers and agricultural extension workers. A farmer could use a smartphone to photograph a suspected diseased coffee plant and receive an AI-generated preliminary classification together with relevant information. The system would not replace agricultural experts; rather, it would serve as an accessible support tool that could help users recognize possible disease earlier and seek appropriate professional advice when necessary.

In the longer term, the research could contribute to greater use of digital technologies in agricultural extension, improved awareness of coffee diseases, and the development of locally relevant AI applications for agriculture.

9. Innovation and Research Contribution

The main contribution of this research is not simply the application of deep learning to coffee disease images. Similar approaches have already been investigated elsewhere. The proposed study focuses on combining several elements that are important for local application:

local field images + deep learning + smartphone deployment + agricultural expert validation + Afaan Oromoo information + farmer/extension-worker evaluation.

10. Multidisciplinary Participation

The research will require collaboration among several fields:

- **Computer Science and Engineering/Information Technology:** artificial intelligence, computer vision, deep learning, mobile application development, and system integration.
- **Agricultural (Horticulture/Plant Science):** coffee production, identification and validation of coffee diseases and interpretation of agricultural recommendations.
- **Statistics/Data Science:** sampling, experimental design, dataset analysis, and model evaluation.
- **Agricultural Extension:** farmer participation, field implementation, and user evaluation.

This multidisciplinary structure is consistent with the university's emphasis on applied research addressing community problems.

11. Estimated Two-Year Research Budget

Table: Estimated budget

No.	Budget item	Estimated ETB
1	Field data collection and transportation	90,000
2	Research assistants/data collectors	80,000
3	Smartphone/camera and field accessories	30,000
4	Computing/GPU, storage and related equipment	100,000
5	Software, hosting, data storage and technical services	60,000
6	Training, expert consultation and workshops	40,000
7	Farmer/extension-worker field evaluation	40,000
8	Communication, printing and research materials	20,000
9	Data management and statistical analysis	20,000
10	Contingency/operational expenses	30,000
	TOTAL	**510,000**

12. Conclusion

Coffee disease management remains a significant challenge for coffee producing communities, particularly in Western Oromia. This research proposes the development of a locally evaluated mobile system that uses deep learning and smartphone-captured images to detect coffee diseases and provide decision support. By integrating computer science and engineering, agriculture, statistics, agricultural extension, and local-language expertise, the study aims to address a critical agricultural problem through artificial intelligence. The expected outcomes include a tested AI-based prototype and locally generated knowledge that can contribute to improved coffee disease management, future digital agricultural services, and further research in coffee-producing communities.

13. Expected Number of Publications

Two (2)